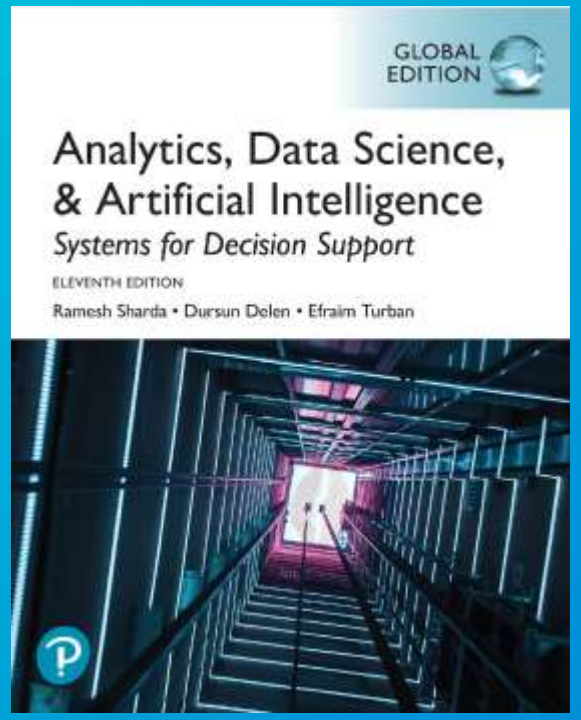
MANAGEMENT SUPPORT SYSTEMS
(415 MIS-3)
CHAPTER – 2



Chapter 2: Artificial Intelligence Concepts, Drivers, Major Technologies, and Business Applications

Analytics, Data Science, & Artificial Intelligence Systems For Decision Support,11ed, Global Edition

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2.2 Introduction To Artificial Intelligence

- Definition of AI
- Major Characteristics of AI Machines
- Major Elements of AI
- AI Applications
- Major Goals of AI
- Drivers of AI
- Benefits of AI
- Some Limitations of AI Machines
- Three Flavors of AI Decisions
- Artificial Brain

What Is Artificial Intelligence (AI)?

- The **major goal** of AI is to create intelligent machines that can do tasks currently done by people.
- **AI tasks** include
  - reasoning,
  - thinking,
  - learning, and
  - problem solving.
- **AI can also be defined as:**
  - Technology that can learn to do things better over time.
  - Technology that can understand human language.
  - Technology that can answer questions.

The Major Benefits of AI

- Significant reduction in the cost of performing work. This reduction continues over time while the cost of doing the same work manually increases with time.
- Work can be performed much faster.
- Work is consistent in general, more consistent than human work.
- Increased productivity and profitability as well as a competitive advantage are the major drivers of AI.

Definition of AI

- Artificial intelligence has several definitions that is concerned with two basic ideas:
  - The study of human thought processes (to understand what intelligence is)
  - The representation and duplication of those thought processes in machines (e.g., computers, robots)

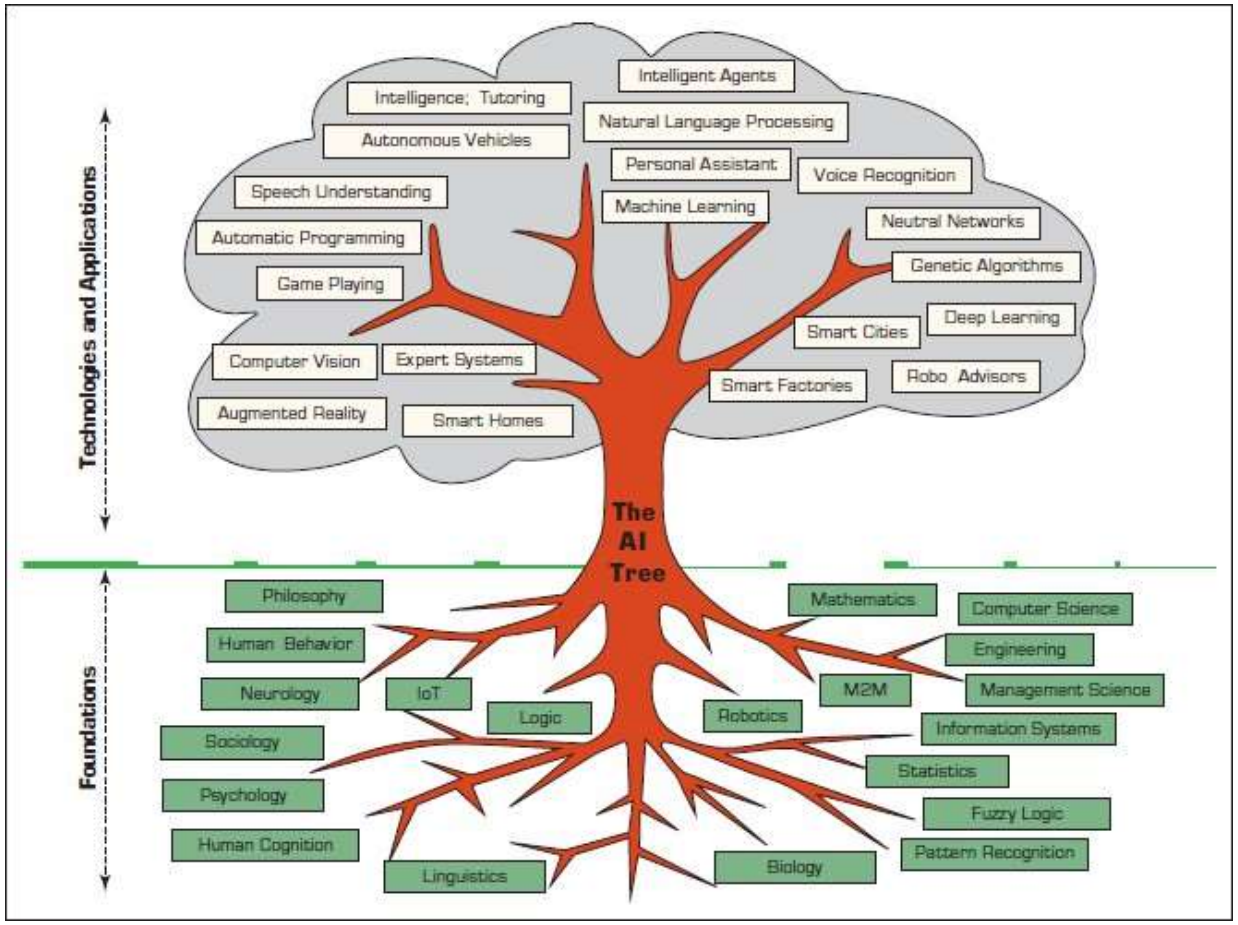
- Another definition of AI is "the capabilities of a machine to imitate intelligent human behavior"

Major Characteristics of AI Machines

- There is an increasing trend to make computers “smarter”.
  - Web 3.0 enables computerized systems that exhibit more intelligence than Web 2.0.
- Several applications are already based on multiple AI techniques.
  - Machine translation of languages is helping people who speak different languages to collaborate in real time as well as to buy online products that are advertised in different languages.

Major Elements of AI

- AI components can be divided into **two** groups: Foundations, and Technologies & Applications.



AI Applications

Smart or intelligent applications include:

- Machines to answer customers' questions asked in natural languages
- Knowledge-based systems which can provide advice, assist people to make decisions, and even make decisions on their own
- Automatic generating of online purchasing orders and arranging fulfillment of orders placed online.
- Shipping prices are determined automatically based on the dimensions, weight, and packaging.

Major Goals of AI

- The overall **goal of AI** is to create intelligent machines that are capable of executing a variety of tasks currently done by people.
- **AI machines** should be able to reason, think abstractly, plan, solve problems, and learn.
- Some specific goals are to:
  - Perceive and properly react to changes in the environment that influence specific business processes and operations
  - Introduce creativity in business processes and decision making

Drivers of AI

The use of AI has been driven by the following:

- People's interest in smart machines and artificial brains
- The low cost of AI applications versus the high cost of manual labor (doing the same work)
- The desire of large tech companies to capture competitive advantage and market share of the AI market and their willingness to invest billions of dollars in AI
- The pressure on management to increase productivity and speed
- The availability of quality data contributing to the progress of AI
- The increasing functionalities and reduced cost of computers in general
- The development of new technologies, particularly cloud computing

Benefits of AI

- AI has the ability to complete certain tasks faster than humans.
- The consistency of AI work. AI machines do not make mistakes, stop, or sleep.
- AI systems allow for continuous improvement projects.
- AI can be used for predictive analysis via its capability of pattern recognition.
- AI can manage delays and blockages in business processes.
- AI machines can work autonomously or be assistants to humans.
- AI machines can learn, improve its performance, and work in hazardous environments.
- AI machines can facilitate innovations by human (i.e., support research and development)
- AI excels in fraud detection and in security facilitations.
- AI can free employees to work on more complex and productive jobs.
- AI can solve difficult problems that previously were unsolved.

Some Limitations of AI Machines

The following are the **major limitations** of AI machines:

- Lack human touch and feel
- Lack attention to non-task surroundings
- Can lead people to rely on AI machines (e.g., people may stop to think on their own)
- Can be programmed to create destruction
- Can cause many people to lose their jobs
- Can start to think by themselves, diminishing with time. However, risks exist. Therefore, it is necessary to properly causing significant damage

- Some of the limitations are diminishing with time. However, risks exist. Therefore, it is necessary to improve AI development and minimize the risks.

The Three Flavors of AI Decisions

The capabilities of AI systems can be divided into three levels

- **Assisted Intelligence:** This is equivalent mostly to the weak AI, which works only in narrow domains. It requires clearly defined inputs and outputs. Examples are some monitoring systems and low-level virtual personal assistants.

- **Autonomous AI:** These systems are in the realm of the strong AI but in a narrow domain. Eventually, the computer will take over as very narrow expert and have absolute decision-making power.

- **Augmented Intelligence:** Most of the existing AI applications are between assisted and autonomous, which is referred to as augmented intelligence. The technology focuses on augmenting computer abilities to extend human cognitive abilities, resulting in high performance

Artificial Brain

- The **artificial brain** is a machine that is desired to be as intelligent, creative, and self-aware as humans. To date, no one has created such a machine.
- The following are some differences between traditional and augmented AI:
  - Augmented machines extend rather than replace human decision making
  - Augmentation excels in solving complex human and industry problems in specific domains in contrast with strong, general AI.
  - In contrast with a “black box” model of some AI and analytics, augmented intelligence provides insights and recommendations, including explanations.

2.3 Human and Computer Intelligence

A. What Is Intelligence?

B. How Intelligent Is AI?

C. Measuring AI

What Is Intelligence?

- **Intelligence** is a broad term measured by an IQ test.

- To understand what artificial intelligence is, it is useful to first examine those abilities that are considered signs of human intelligence:
  - Learning or understanding from experience
  - Making sense out of ambiguous, incomplete, or even contradictory messages and information
  - Responding quickly and successfully to a new situation
  - Understanding/inferring in a rational way, and solving problems
  - Applying knowledge to manipulate environments and situations
  - Recognizing & judging the relative importance of elements in a situation

How Intelligent Is AI?

- AI machines have demonstrated superiority over humans in playing complex games such as chess, Jeopardy!, and Go by defeating the world's best players.
- Despite this many AI applications still show significantly less intelligence than humans.

TABLE 2.1 Artificial Intelligence versus Human Intelligence

Area	AI	Human
Execution	Very fast	Can be slow
Emotions	Not yet	Can be positive or negative
Computation speed	Very fast	Slow, may have trouble
Imagination	Only what is programmed for	Can expand existing knowledge
Answers to questions	What is in the program	Can be innovative
Flexibility	Rigid	Large, flexible
Foundation	A binary code	Five senses
Consistency	High	Variable, can be poor
Process	As modeled	Cognitive
Form	Numbers	Signals
Memory	Built in, or accessed in the cloud	Use of content and scheme memory

Measuring AI

- The Turing Test is a well-known attempt to measure the intelligence level of AI machines.
- It **aims** to determine whether a computer exhibits intelligent behavior. A computer can be considered smart only when a human interviewer asking the same questions to both an unseen human and an unseen computer cannot determine which is which.
- To pass the **Turing Test**, a computer needs to be able to understand a human language (NLP), to possess human intelligence (e.g., have a knowledge base), to reason using its stored knowledge, and to be able to learn from its experiences (machine learning).

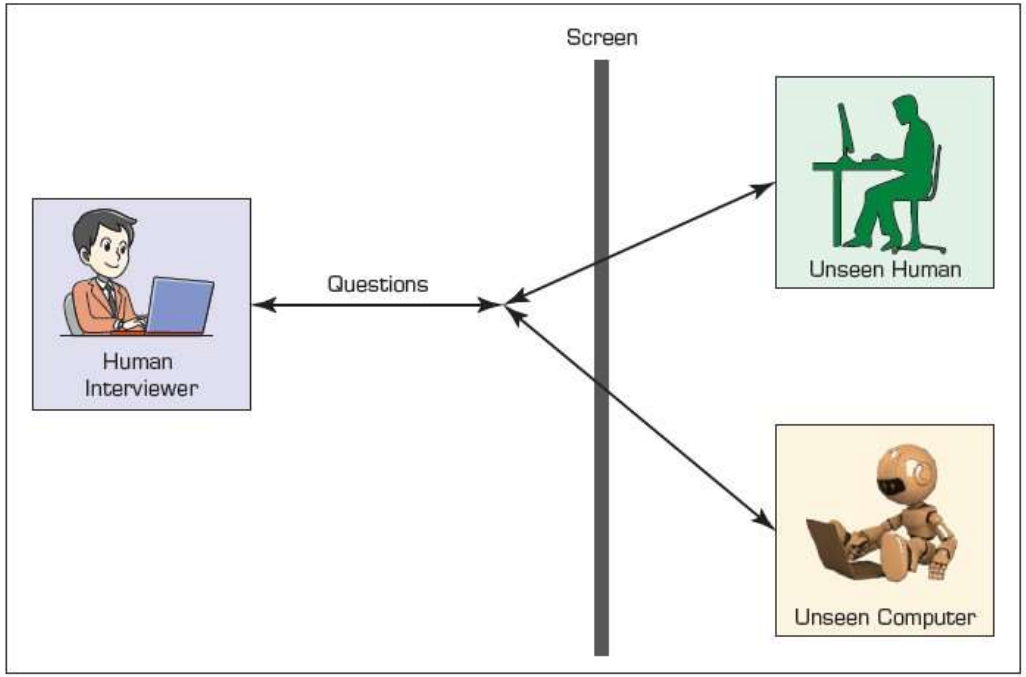


FIGURE 2.3 A Pictorial Representation of the Turing Test

2.4 Major AI Technologies And Some Derivatives

- Intelligent Agents
- Machine Learning
- Deep Learning
- Machine and Computer Vision
- Robotic Systems
- Natural Language Processing
- Knowledge and Expert Systems and Recommenders
- Chatbots
- Emerging AI Technologies

Intelligent Agents

- An **intelligent agent (IA)** is a small computer software program that observes and acts upon changes in its environment by running specific tasks autonomously.

- An IA directs an agent's activities to achieve specific goals related to the changes in the surrounding environment.

- IAs have the ability to learn by using their knowledge

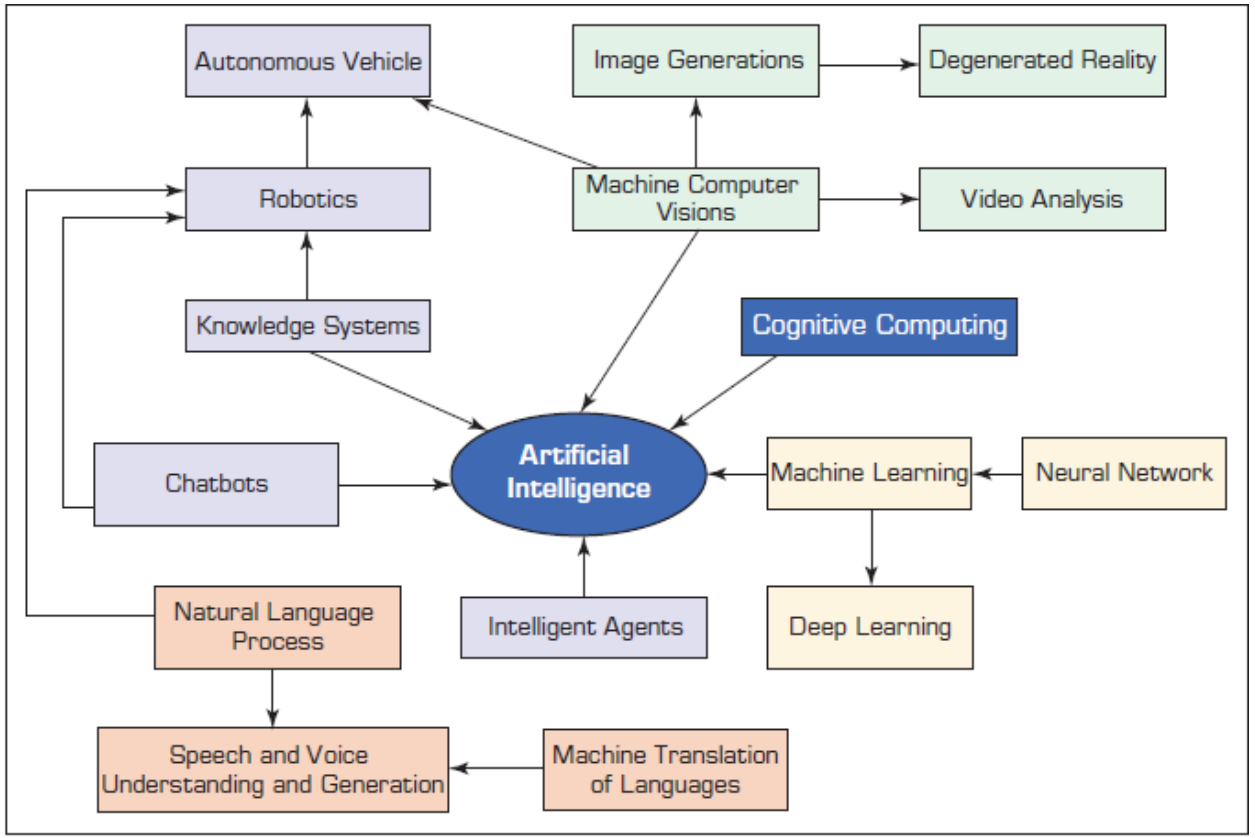


FIGURE 2.4 The Major AI Technologies

Intelligent Agents (cont.)

- IAs are tools for **overcoming Internet** information overload and making computers viable decision support tools.

  - **Example 1:** An example of an intelligent software agent is a virus detection program. It resides in a computer, scans incoming data, and removes viruses automatically while learning to detect new virus types and detection methods.

  - **Example 2:** Allstate Business Insurance uses an intelligent agent to reduce call center traffic and provide human insurance agents during the rate-quoting process with business customers.

Machine Learning

- **Machine Learning** (ML) is a discipline concerned with design & development of algorithms that allow computers to learn based on incoming data.
- ML allows computer systems to monitor and sense their environment, so that the machines can adjust their behavior to deal with the changes
- ML scientists teach computers to identify patterns and make connections by showing the machines a large volume of examples and related data.
- **ML used** for predicting, recognizing patterns, & supporting decision makers.
- **ML example** situations to a machine-learning program it can find hidden elements. An example is computers detecting credit card fraud.
- **ML applications** are expanding due to the availability of Big Data sources, especially those provided by the IoT.

Deep Learning

- One subset of machine learning is called deep learning; a technology tries to mimic how the human brain works.
- **Deep learning** (DL) uses artificial neural technology and deals with complex applications that regular machine learning and AI technologies can not handle.
- DL delivers systems that think and learn, enabling self-direction. This allows DL to tackle previously unsolvable problems using powerful learning algorithms.
- For example, DL is a key technology in autonomous vehicles by helping to interpret road signs and road obstacles.
- DL is mostly useful in real-time interactive applications in the areas of machine vision, scene recognition, robotics, and speech and voice processing.

Machine and Computer Vision

- **Machine vision** includes “technology and methods used to provide imaging-based automated inspection and analysis for applications such as robot guidance, process control, autonomous vehicles, and inspection.”
- **Computer vision** “is an interdisciplinary field that deals with how computers can be made for gaining high-level understanding from digital images or videos. From the perspective of engineering, it seeks to automate tasks that the human visual system can do.”
- Both computer vision and machine vision automate many human tasks (e.g., inspection). However, Machine vision is treated more as an engineering subfield, while computer vision belongs to the computer science area.

Machine and Computer Vision

- The two technologies are combined with image processing that facilitates complex applications, such as visual quality control.
- Applied area of machine vision is scene recognition, done by computer vision.
- Video analytics is a derivative application of computer vision, where techniques are applied to videos to enable pattern recognition and identify events.
- Example of Applications:
  - The machine vision wood identification project developed a prototype machine vision system for wood identification to help identify illegal logging.
  - AI computer vision mixed with deep learning identifies illegal animal poachers.
  - Facial recognition that employ smart glasses to identify potential suspects.

Robotic Systems

- A robot is a device guided by a program to perform manual/mental tasks.
- An "intelligent" robot has a sensory apparatus (e.g., camera) to collects information about the surroundings and can respond to the changes in the environment.
- Autonomous robots (programmed to do tasks completely on their own, even repair themselves), are equipped with AI intelligent agents.
- Example: Walmart Is Using Robots to Properly Stock Shelves
  - In Walmart, 2-foot-tall robots use cameras/sensors to scan the shelves for misplaced, missing, or mispriced items. The results are transmitted to humans who take corrective actions. The robots carry out their tasks faster and frequently more accurately than humans.

Natural Language Processing

- **Natural language processing** (NLP) allows users to communicate with a computer in their native language. NLP includes two subfields:
  - NLP that investigates methods of enabling computers to comprehend instructions or queries provided in English or other human languages.
  - NLP generation that strives to have computers produce ordinary spoken language so that people can understand the computers more easily.
- **Speech/Voice Understanding**: recognition & comprehension of spoken languages by a computer. This has been adopted in automated call centers.
- **Machine Translation Of Languages:** uses computer programs to translate words and sentences from one language to another.

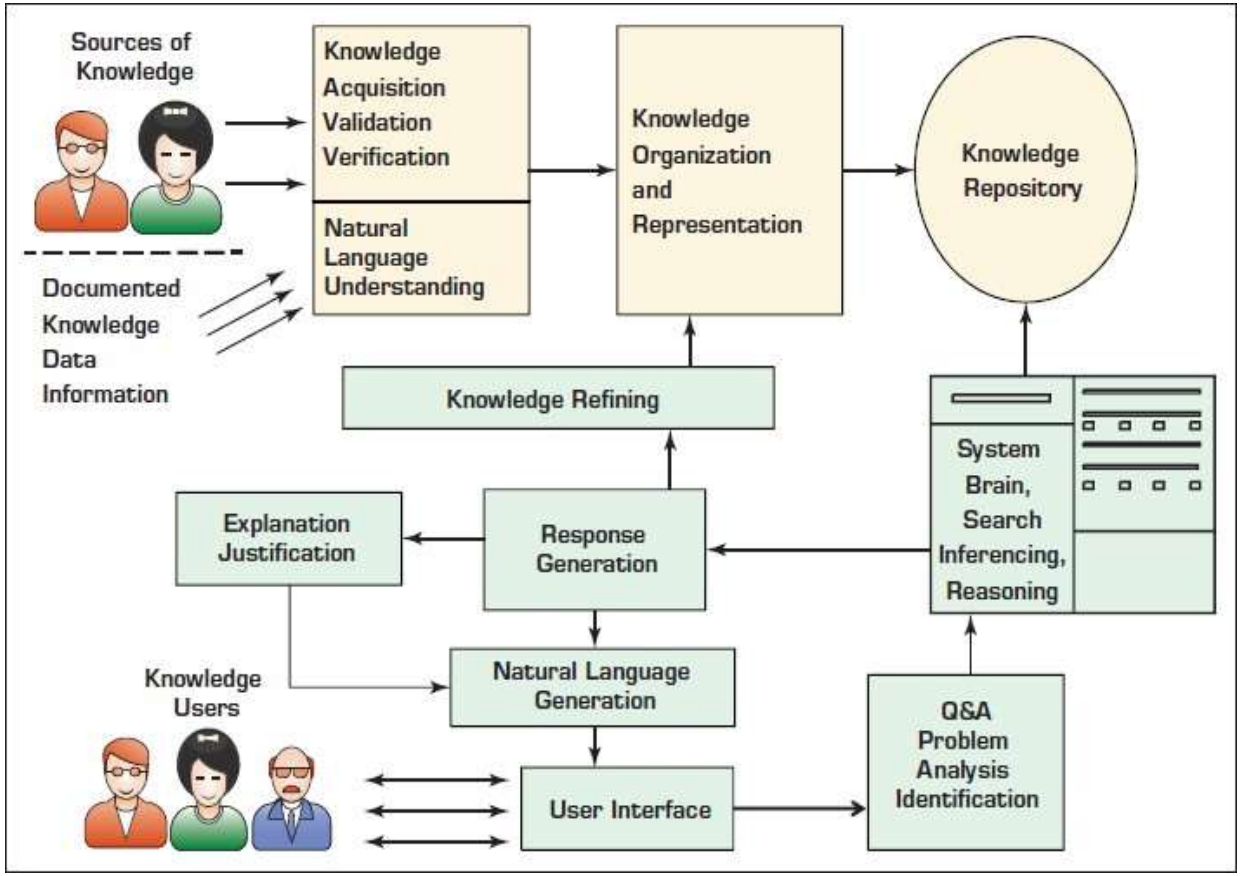
Knowledge and Expert Systems and Recommenders

- These systems are computer programs that store knowledge, which their applications use to generate expert advice and/or perform problem solving.
- **Knowledge-based expert** systems help people to verify information and make certain types of automated routine decisions.
- **Recommendation systems** are knowledge-based systems that make recommendations to people. Another knowledge system is chatbots.
- **Knowledge Sources And Acquisition For Intelligent:** Intelligent systems must gain knowledge through knowledge acquisition.
- Knowledge acquisition includes extracting and structuring knowledge from data, and then experts may be used to verify it.

Knowledge and Expert Systems and Recommenders

- **Knowledge Representation:** the organization and storage of acquired knowledge. The method of representation depends on the use of knowledge, reasoning from knowledge, user interaction with the knowledge, and more.

- **Reasoning From Knowledge:** processing a users' requests and gives answers. The difference among intelligent technologies is their type of reasoning.



Chatbots

- Robots come in several shapes and types, of which is a chatbot. A **chatbot** is a conversional robot that is used for chatting with people.
- Depending on the purpose of the chat, which can be done in writing or by voice, bots can be in the form of intelligent agents that retrieve information or personal assistants that provide advice.
- In either case, chatbots are usually equipped with NLP that enables conversations in natural human languages rather than in a programmed computer language.

Emerging AI Technologies

- Several new AI technologies are emerging. Here are a few examples:
  - **Effective computing.** Technologies that detect emotional conditions of people and suggest how to deal with discovered problems
  - **Biometric analysis**. Technologies that verify an identity based on unique biological traits that are compared to stored ones (e.g., facial recognition).
- **Cognitive Computing:** The application of knowledge derived from cognitive science so that computers can exhibit and/or support decision-making and problem-solving capabilities.
- **Augmented Reality:** Augmented reality (AR) refers to the real time integration of digital information with a users environment (mostly vision and sound). The technology provides a real-world interactive experience with the environment.

2.5 AI Support For Decision Making

- Issues and Factors in Using AI in Decision Making
- AI Support of the Decision-Making Process
- Automated Decision Making

Issues and Factors in Using AI in Decision Making

These factors determine the justification of AI usage and its chance of success:

- The nature of the decision (E.g., routine decisions = likely automated)
- The method of support, what technologies are used

- **Cost-benefit & Risk Analyses:** necessary for large-scale decisions, but hard to compute with AI models due to difficulties in measuring costs, risks, & benefits.
- **Using Business Rule:** AI systems can be based on business rules, whose quality determines that of the automated.
- **AI Algorithms:** are the basis for automated decisions & decision support.
- **Speed:** Decision automation is dependent on the speed decisions needs to be made. Some decisions takes too much time to get all the relevant input data.

AI Support of the Decision-Making Process

AI support can be applied to the various **steps** of the **decision-making process**:

1. **Problem Identification:** collecting data through technology that can be used by AI algorithms. Performance levels of machines are compared to standards, and trend analysis can point to opportunities.

2. **Generating/Finding Alternative Solutions:** matching problem characteristics with best practice, or proven solutions stored in databases. Tools such as case-based reasoning and neural computing are for this purpose.

3. **Selecting a Solution:** evaluate proposed solutions, predict future impacts, asses chance of success, or predict a reply to actions taken by a competitor.

4. **Implementing the Solutions:** demonstrates the superiority of proposals and to assess resistance to changes.

Automated Decision Making

- The process of automated decision making starts with knowledge acquisition and creation of a knowledge repository.
- The system generates and submits responses to user's questions. Solutions are evaluated to improve the knowledge repository, and complex situations are forwarded to humans.
- Companies use automated decision making for both their external operations (e.g., sales) and internal operations (e.g., resource allocation)

  Example: Supporting Nurses' Diagnosis Decisions

- Researchers used AI tools to conduct data mining to predict the probable success of automated nursing diagnoses based on patient characteristics.

2.6 AI Applications in Accounting

- AI in Accounting: An Overview
- Accounting Applications in Small Firms
- Job of Accountants

AI in Accounting: An Overview

- The CEO of an Accounting Software for businesses noticed trends among professional accountants: an increase in their use of AI, including bots in professional routines.

- It was observed that the major drivers for this are perceived savings in time and money and increased accuracy and productivity. The adoption has been rapid and it has been followed by significant improvements.

- An example is the execution of compliance procedures. For instance, machine learning is used for detecting anomalous data (e.g., fraudulent invoices).

Accounting Applications in Small Firms

- Crowe Horwath accounting firm uses AI to solve **healthcare billing problems**.
- Other applications like analyzing real estate contracts & risk analysis are used.
- A comprehensive study of AI in accounting report the advantage of AI as:
  - Provides cheaper, better data to support decision making
  - Generating insight from data analysis
  - Freeing time of accountants to concentrate on problem solving
- The report points to the use of the following:
  - Machine learning for detecting fraud and predicting fraudulent activities
  - Knowledge-based systems for verifying of accounting tasks
  - Deep learning to analyze unstructured data, such as contracts and e-mails

Job of Accountants

- AI and analytics will automate many routine tasks done today by accountants, many of whom may lose their jobs
- On the other hand, accountants will need to manage the used AI-based accounting systems
- Accountants will need to drive AI innovation in order to succeed or possibly even survive in the business

2.7 AI Applications in Financial Services

- AI Activities in Financial Services
- AI in Banking: An Overview
- Illustrative AI Applications in Banking
- Insurance Services

AI Activities in Financial Services

These activities may be found across various types of financial services:

- Extreme personalization (e.g., chatbots, assistants, robo investment advisors)
- Shifting customer behavior both online and in brick-and-mortar branches
- Facilitating trust in digital identity
- Revolutionizing payments
- Sharing economic activities (e.g., person-to-person loans)
- Offering financial services 24/7 and globally (connecting the world)

AI in Banking: An Overview

- AI is transforming the **banking industry** in areas of AI IT, finance and accounting, marketing & sales, human resource management, customer service, and operations.

The following are some points on the relation between AI and banking:

- AI technologies in banking include many analytical tools.
- These technologies help banks improve both their front-office and back-office operations.
- Major activities are the use of chatbots to improve customer service and communicating with customers, and robo advising is used by some financial institutions.
- Facial recognition is used for safer online banking.
- Advanced analytics helps customers with investment decisions.
- AI algorithms help banks identify and block fraudulent activities (E.g., money laundering).
- AI algorithms can help in assessing the creditworthiness of loan applicants.

Illustrative AI Applications in Banking

The following are applications of AI in banking:

- **AI machines** that step up employee surveillance, and prevent illegal activities
- **Applications** that ensures individuals pay what they owe, and tries to lower tax bills.
- **Bots** that assist staff members to find the appropriate answers to queries in real time.
- **Robots** that analyze customer facial expression and determine their nationality. Then determines and selects the matching language of the customers.

Example: Banks Manage Compliance and Supports Decision Making

- Banks must examine huge amounts of data to ensure compliance with government regulations. Thus, a set of cognitive tools dealing with regulatory compliance was created.
- One of the tools deals with financial crimes, flagging suspicious transactions and fraud. The second tool monitors compliance, and the third deals with large volumes of data.

Insurance Services

- Advancements in AI are improving issuing policies and handling claims. Incoming claims are analyzed by AI, and are sent to appropriate available adjusters.
- The AI software can help in data collection and analysis and in data mining old claims.
- The machines provide speed, accuracy, and efficiency in performing this process. AI improves analysis results and enhance customer experience.

Example: Metromile Uses AI in Claim Processing

- Metromile is a vehicle insurance company, using the pay-per-mile model. It uses an AI-based program to automate accident data, process claims, and pay customer claims.
- The AI bot simulates the accidents' major points and makes a verification based on decision rules; authorization for payments provides for successful verification.
- Only complex cases are sent to investigation by human processors.

2.8 AI in Human Resource Management (HRM)

- AI in HRM: An Overview
- AI in Onboarding
- Introducing AI to HRM Operations

AI in HRM: An Overview

The following are ways in which AI can transform HRM in recruiting:

1. Reducing human bias & Improving relationships with current employees
2. Increasing efficiency, productivity, and insight in evaluating candidates

The impact of AI into the following areas:

- **Recruitment (Talent Acquisition):** AI improves recruiting by helping recruiters and job seekers by using AI algorithms to suggest matches. This removes biases and prejudice.
- **AI Facilities Training:** AI methods can be used to facilitate learning. For example, chatbots can be used as a source of knowledge to answer learners' queries.
- **AI Supports Performance Analysis (Evaluation):** AI enables HR management to conduct performance analysis and measure the performance of employees. The performance is compared to objectives, which are provided to employees and teams.
- **AI Use In Retention And Attrition Detection:** Machine learning can be used to detect reasons why employees leave companies by identifying influencing patterns.

AI in Onboarding

- The HR department needs help introducing new employees to the organizational culture and operating processes.
- Some new employees require much attention and AI can help prepare customized onboarding paths that are best for the newcomers.
- Results showed that those employees supported by AI-based plans tend to stay longer in organizations
- The use of chatbots in HRM is increasing rapidly. Their ability to provide current information to employees anytime is a major reason.

Introducing AI to HRM Operations

- Introducing AI to HRM operations is like introducing AI to functional areas.

The following are suggested activities of AI in this area:

- Experiment with a variety of chatbots
- Develop a team approach involving other functional areas
- Properly plan a technology roadmap for both the short and long term, including shared vision with other functional areas
- Identify new roles & modifications to existing roles in transformed environments
- Train and educate the HRM team to understand AI and gain expertise in it

2.9 AI in Marketing, Advertising, and CRM

- Overview of Major Applications
- Customer Experiences and CRM
- Other Uses of AI in Marketing

Overview of Major Applications

The following are examples of AI in marketing:

1. Product & personal recommendations
2. Smart search engines
3. Fraud and data breaches detection
4. Social semantics
5. Website design
6. Producer pricing
7. Predictive customer service
8. Ad targeting
9. Speech/Image recognition
10. Language translation
11. Customer Segmentation
12. Sales forecasting
13. Content generation
14. Using bots, robo advisors, etc.

Customer Experiences and CRM

A major impact of AI technologies is changing customer experiences through:

1. Use NLP for creating user documentation. This improves customer–machine dialogue.
2. Use visual categorization to organize images (E.g., IBM's Visual Recognition & Clarifai)
3. Provide personalized and segmented services by analyzing customer data. This includes improving shopping experience and CRM.

Example: Salesforce's AI Einstein

- Salesforce Einstein is an AI set of technologies used for enhancing customer interactions and supporting sales. The system delivers dynamic sales dashboards to sales reps, tracks performance and manages teamwork by using sales analytics. The AI product also can provide predictions and recommendations.
- Einstein's automatically prioritized sales leads make sales reps more productive when dealing with sales leads and potential opportunities. The sales reps also get insights about customers' sentiments, competitors' involvement, and other information.

Other Uses of AI in Marketing

The following show the diversity of AI technologies used in marketing:

- **Bots** can mimic the expertise of in-store sales people, and make shopping easier.
- It provides lead generation. AI can help generate sales leads by analyzing customers' data. The program can generate predictions. Insights can be generated by intelligent analytics.
- It can increase customer loyalty using personalization.
- It can improve the sales pipeline. Robots can convert unknown visitors into customers using the **following three stages**:
    1. Prepare a list of target customers in the database
    2. Send information, ads, videos, and so on to prospects on the list created earlier
    3. Provide the company sales department with a list of leads that successfully convert potential customers to buyers.

2.10 AI Applications in Production-Operation

- AI in Manufacturing
- Implementation Model
- Intelligent Factories
- Logistics and Transportation

AI in Manufacturing

- To handle increasing labor costs, customers' requirement changes, increased competition, etc. companies use high levels of automation/digitization.
- Companies need to be more agile, and react quicker and more effectively.
- They also need to be more efficient and improve customers' (organizations' and individuals') experiences.
- To achieve these goals, they need to automate processes and make use of AI and other cutting-edge technologies.

Implementation Model

The five-component model to use intelligent technologies includes:

1. Streamlining processes, including minimizing waste, redesigning processes, and using business process management (BPM)
2. Outsourcing certain business processes, including going offshore
3. Using intelligence in decision making by deploying AI and analytics
4. Replacing human tasks with intelligent automation
5. Digitizing customers' experiences

- Companies use robots for complex tasks, where mental tasks are automated. AI and sensors allow supporting or automating production decisions in real time.

Intelligent Factories

- Companies employ smart or intelligent factories, who use complex software and sensors.
    - An example of a lead supplier is General Electric, which provides software such as OEE Performance Analyzer and Production Execution Supervisor. .
- Many small vendors are specializing in different aspects of AI for manufacturing.
    - For example, BellHawk Systems Corporation, provides services to small companies, specializes in real-time operations tracking.

Logistics and Transportation

- AI and intelligent robots are used extensively in corporate logistics and internal and external transportation, as well as in supply chain management.
- Example: DHL Supply Chain
  - DHL is a global delivery company developing innovative logistics and transportation business models, mostly with AI, IoT, and machine learning.
  - Several of the IoT projects are linked to machine learning, in the areas of sensors, communication, device management, security, and analysis.
  - Tagging and tracking items using Radio Frequency Identification (RFID) and Quick Response (QR) code allow for item tracking along the supply chain.

Reference

- Ramesh Sharda, Dursun Delen, Efraim Turban; Analytics, Data Science, & Artificial Intelligence: Systems For Decision Support, 11ed, Global Edition, Pearson Education Limited, 2021, ISBN 10: 1 - 2 92-34155-6, ISBN 13: 978-1-292-34155-2

Chapter 2: Artificial Intelligence Concepts, Drivers, Major Technologies, and Business Applications

Recommended Reading

- **AI-powered decision support systems, what are they?**
  https://blog.pwc.lu/ai-powered-decision-support-systems-what-are-they/

Recommended Videos

- **Artificial intelligence and decision-making (by Thorbjørn Knudsen)**
  https://www.youtube.com/watch?v=Sujww4njwE4